

BODY SYSTEMS

Science Eight – Unit I

BODY SYSTEMS

All the body systems
work together to
allow us to function

This unit looks at
connections between
systems and how our
choices influence our
body systems

STATEMENT OF INQUIRY & CONCEPTS

STATEMENT OF INQUIRY

- By understanding how our body **systems function**, people can learn to make decisions for **balanced** and healthy lifestyles

CONCEPTS

Key Concept:

- Systems

Related Concepts:

- Balance, Function

Global Context Exploration:

- Personal and cultural expression

FACTUAL & CONCEPTUAL QUESTIONS

FACTUAL QUESTIONS

- What are the structures and functions of different body systems?
- How do our body systems work together?
- What physical and chemical changes do body systems undergo?

CONCEPTUAL QUESTIONS

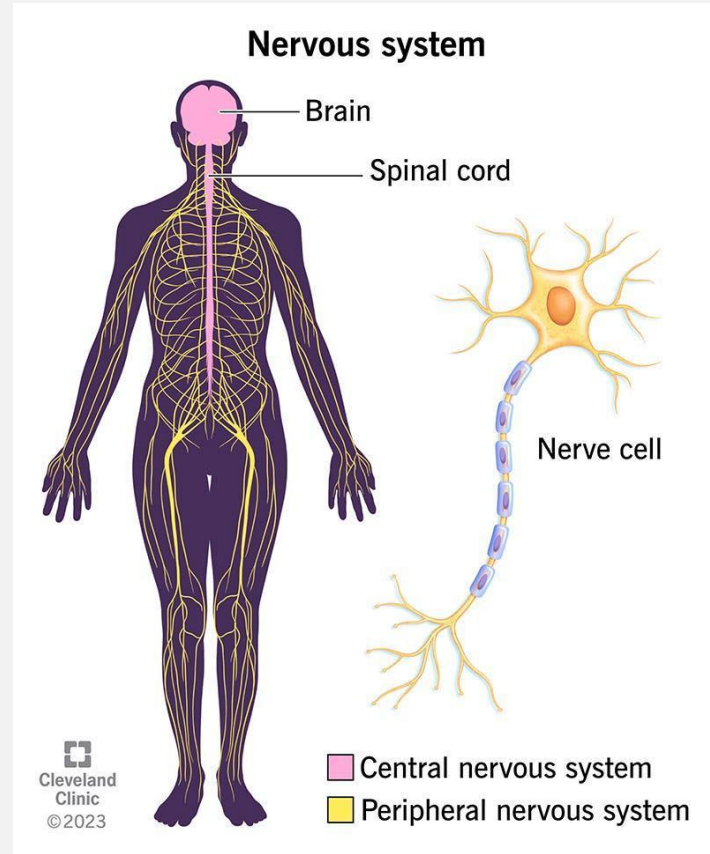
- How can understanding body systems help us make decisions for balanced and healthy lives?
- How can people effectively share information about human body processes?

WHAT ARE BODY SYSTEMS MADE OF?

- **Body systems** are made up of different organs and blood vessels
- The **muscle** found throughout the body is part of the **Muscular System**
- All of our body organs and blood vessels are made of **muscle**; we also have muscles attached to our bones
- **Muscles are** fibrous tissues that can contract and relax to allow for strength and movement
- **Digestive** organs must undergo contractions to move food material along the digestive tract

THE NERVOUS SYSTEM

- The nervous system is made up of the:
- Brain
- Spinal Cord
- Nerves that run all over the body
- The nervous system sends messages in the form of electrical signals between the brain and the body
- These electrical signals (or nerve impulses) tell us what to do



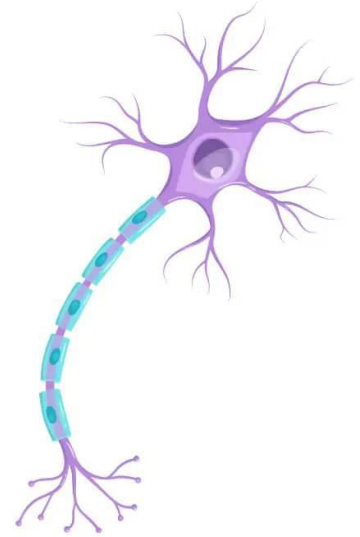
<https://my.clevelandclinic.org/health/body/21202-nervous-system>

WHAT ARE NERVES?

- Nerves are bundles of fibers
- The fibers are made of bundles of cells called **neurons**
- **Neurons** send electrical signals throughout the body that tell our bodies to feel things and do things (ex. You might shiver when it's cold, or your hand might pick up an object)

What is a Neuron?

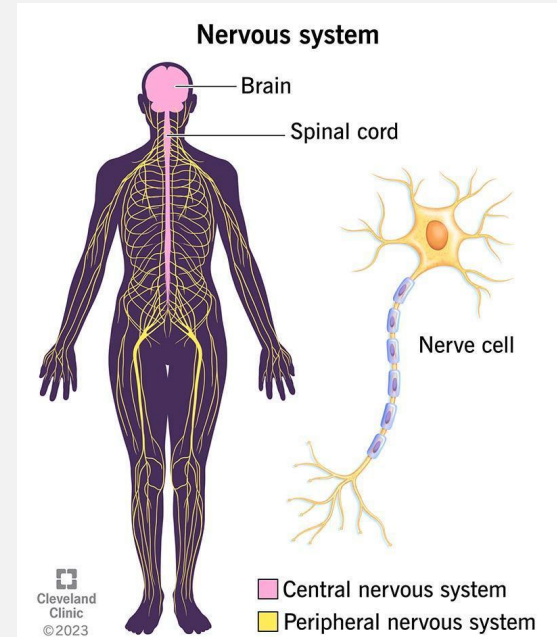
Neurons, which are also referred to as 'nerve cells,' can be best defined as the basic units of your nervous system and brain.



THE NERVOUS SYSTEM

- The brain controls the functions of the body
- The spinal cord runs from the brain down through the back
- The spinal cord contains threadlike nerves that branch out to all organs and body parts (**nervous and all other systems**)
- **Neurons** send messages back and forth from the brain to the nerves throughout the body via the spinal cord (**nervous and all other systems**)
- The messages are in the form of **electrical signals** called **nerve impulses** (see video below on the heart signals)

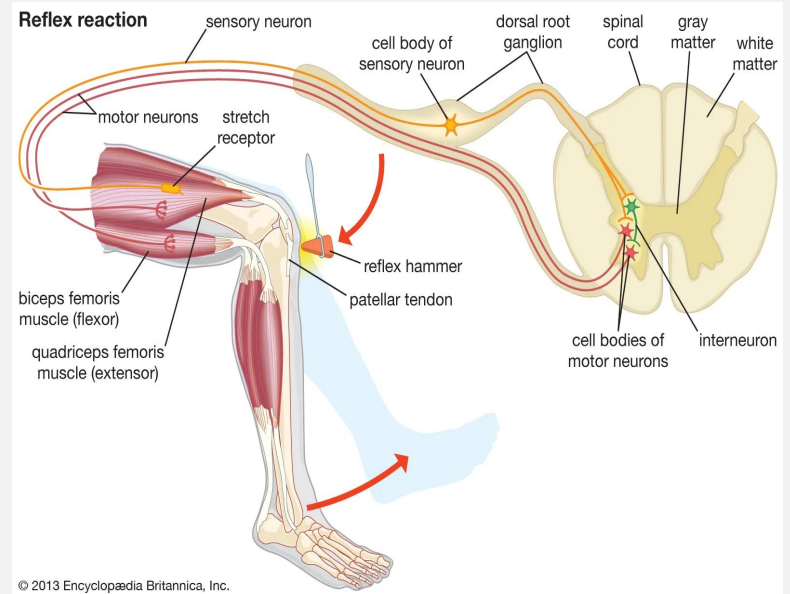
<https://www.youtube.com/watch?v=RYZ4daFwMa8>



<https://my.clevelandclinic.org/health/body/21202-nervous-system>

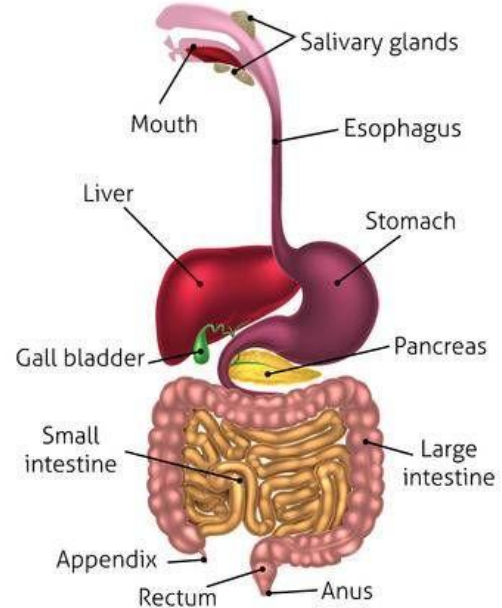
So, why does your leg move when you hit the knee in a certain spot?

- When you hit the knee in the right spot below the knee cap, the leg will move (knee jerk reflex)
- This is because you have hit a sensory receptor which is part of a nerve
- This tells the sensory neuron to send a signal up the spinal cord to the brain where a motor receptor is activated
- This tells the motor neuron to send a message back telling the leg to move



DIGESTIVE SYSTEM

- The **digestive system** processes our food and turns it into nutrients our bodies can use
- **Anatomy:** mouth, teeth, pharynx, esophagus, stomach, liver, small and large intestines, anus, rectum
- **Physical and chemical changes** are used to process food
- **Physical Changes:** Changes that do not make any new substances; the starting materials are retained
- **Chemical Changes:** Changes that produce new substances; the starting materials are used up over the course of the reaction



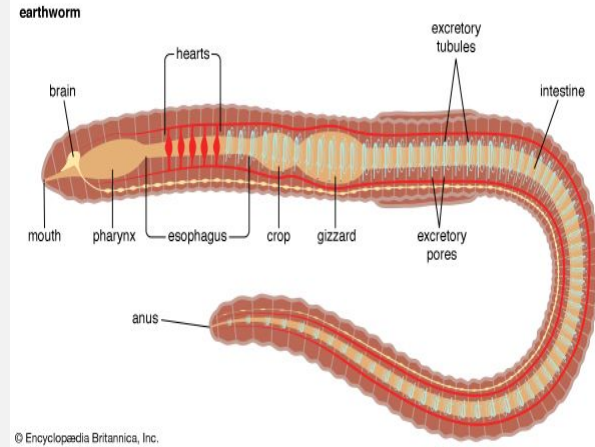
CONNECTIONS: MOVEMENT OF FOOD THROUGH THE DIGESTIVE TRACT

- In humans, food is chewed in the mouth to break it into smaller pieces and mix it with saliva (**physical change**)
- The **chewed food** passes from the mouth to the esophagus via the pharynx
- The esophagus undergoes **peristalsis (muscle contractions)** to push the food down to the stomach (**muscular and digestive**)
- In the stomach, the food mixes with acid and other chemicals causing it to change chemically into an acidic liquid (**Chemical Change**)
- The **acidic liquid** moves from the stomach into the small intestine where all of the nutrients are absorbed into the bloodstream by **diffusion (circulatory and digestive)**
- What is left over after nutrient absorption is waste that passes into the large intestine (or colon). Here the waste fully forms and gets ready to leave the body via the rectum and anus

EARTHWORM DIGESTIVE ANATOMY

- **Mouth:** Body opening on the anterior end; used to ingest and moisten food for easier movement
- **Pharynx:** Cavity that undergoes muscular contractions to move food from the mouth to the esophagus (**digestive and muscular**)
- **Esophagus:** A tube that moves food from the pharynx to the crop
- **Crop:** A soft, stretchy organ that stores food
- **Gizzard:** A hard organ that contracts to physically break down food into smaller bits; it gets help from hard particles in the soil

Digestive systems compared



<https://kids.britannica.com/students/assembly/view/193074>

EARTHWORM DIGESTIVE ANATOMY

- **Intestine:** A long tube where all of the chemical digestion and nutrient absorption occurs; cells lining the intestine make enzymes for chemical digestion

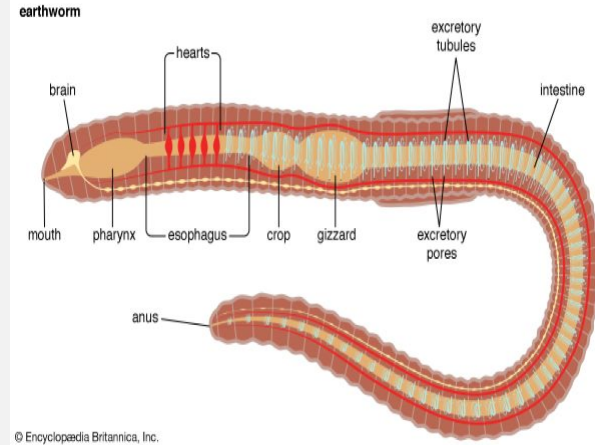
ADAPTATION: Length of the intestine

- A long intestine ensures that the food stays in the intestine long enough so that all of the nutrients can be absorbed from the food

ADAPTATION: Intestinal folds

- Folds increase **surface area** so more nutrients can be absorbed
- **Anus:** The opening at the posterior end of the worm where waste is expelled

Digestive systems compared



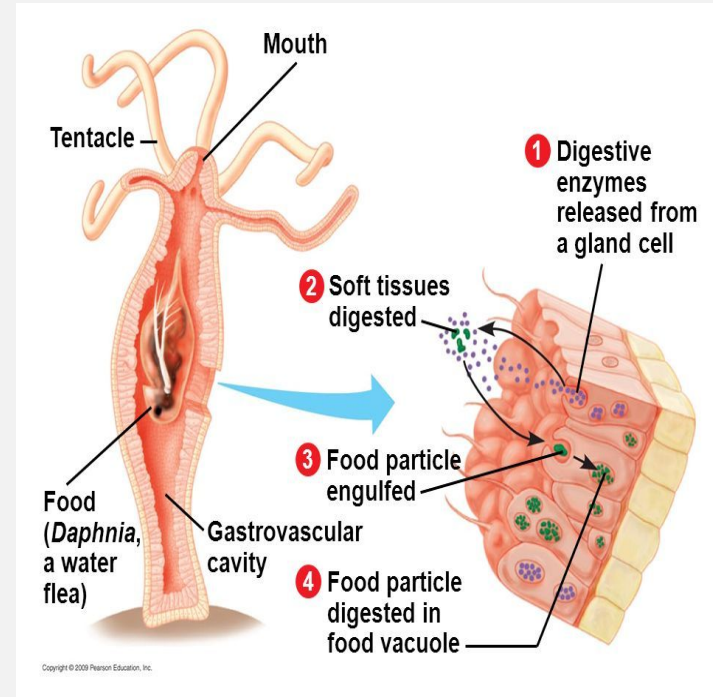
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<https://kids.britannica.com/students/assembly/view/193074>

HYDRA DIGESTION

- **Hydra** are tiny freshwater organisms that can't see or move much
- They detect prey organisms nearby using **sensory cells**
- They grab prey organisms using **tentacles** and push food into the **mouth**
- Food goes directly from the mouth into the **cavity**
- This cavity is lined with **cells** that produce digestive **enzymes** (**chemicals that help with digestion**)

https://www.youtube.com/watch?v=dl_oVns2oa8



<https://socratic.org/questions/59e0d3f37c014943b519be2d>

HYDRA DIGESTION (CONTINUED)

- Enzymes use **diffusion (the involuntary movement of substances from an area of high to low concentration across a membrane)** to pass from the cell into the cavity
- Some enzymes hit the food breaking it into smaller pieces (**physical digestion**)
- Some enzymes hit the smaller food particles changing them into a new material that hydra can use (**chemical digestion**)
- Breaking the food into smaller pieces increases the **surface area** of the food, allowing enzymes to act on more sides making the food easier to chemically digest

BODY SYSTEM CONNECTIONS BETWEEN HUMANS AND OTHER ORGANISMS

BOTH EARTHWORMS AND HUMANS

- Have a mouth, pharynx, and esophagus
- Have a long Intestine allowing more time for nutrient absorption
- Have a folded intestine allowing for more surface area over which nutrients can be absorbed
- Have an anus and produce waste
- Use diffusion for nutrient absorption
- Undergo both physical and chemical digestion
- Produce digestive enzymes
- Have a stretchy organ that holds food (stomach and crop)
- Have blood vessels and a heart

BOTH HYDRA AND HUMANS

- Have a mouth to ingest food
- Undergo physical and chemical digestion
- Produce waste
- Both have an organ that stretches so it can hold more food (human stomach and hydra cavity)
- Both have cells that give off digestive enzymes
- Both have body parts to grab food (hands and tentacles)
- Both use diffusion for nutrient absorption

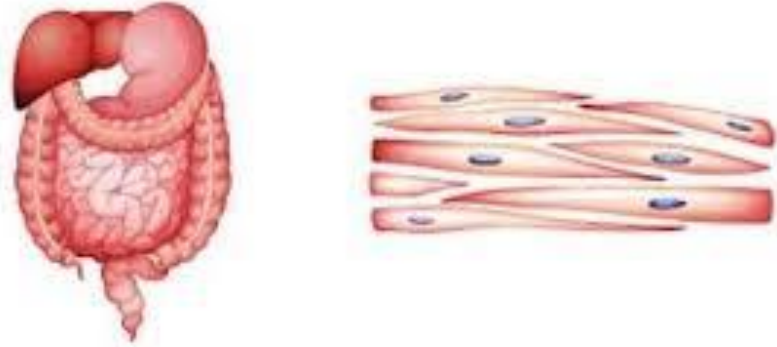
BODY SYSTEM CONNECTIONS BETWEEN HYDRA AND WORMS

- Both have a mouth to ingest food
- Both use physical and chemical digestion
- Both have an organ that can stretch to store more food (cavity and crop)
- Both have cells that produce digestive enzymes
- Both produce waste
- Both use diffusion for nutrient absorption

MUSCULAR AND DIGESTIVE SYSTEM CONNECTIONS

MUSCLES HELP:

- Nutrients **diffuse** into the bloodstream via blood vessels (**digestive and circulatory**)
- Help the esophagus and intestines to undergo contractions to move food along (**digestive**)



RESPIRATORY SYSTEM ANATOMY AND CONNECTIONS

Nose and
mouth

Pharynx

Trachea (or
windpipe)

Bronchi

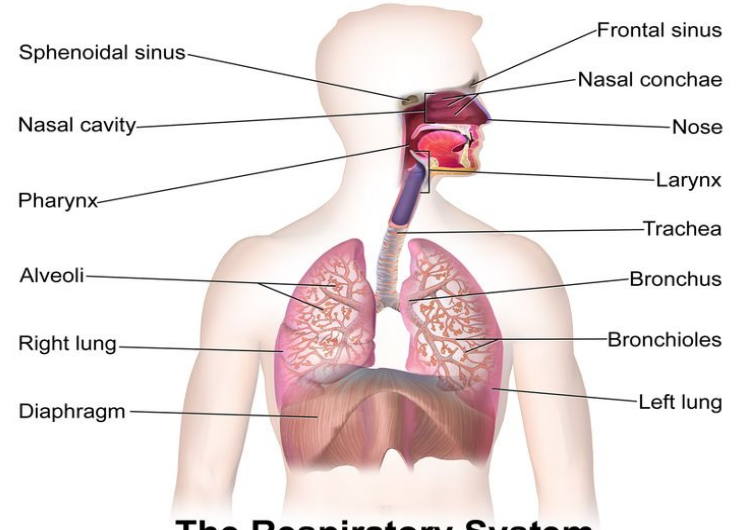
Bronchioles

Alveoli

Diaphragm

Surface
Area

Diffusion



<https://www.vedantu.com/question-answer/a-draw-a-labelled-diagram-of-the-human-class-12-biology-cbse-5f7980b26d1f563a8e1a5486>

MUSCULAR AND RESPIRATORY CONNECTIONS

Some parts of respiratory system are made of **muscle such as the:**

- Diaphragm
- Intercostal muscles (located in between the ribs)
- Walls of the bronchial tubes

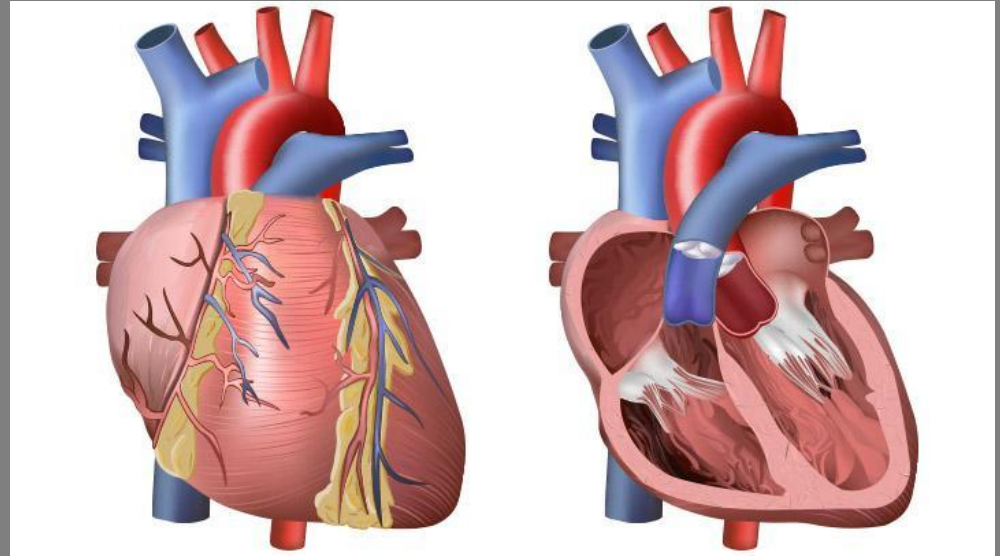
These muscles help with breathing by:

- Moving up and down to allow for the expansion and contraction of the lungs (Diaphragm)
- Help the chest expand and contract when breathing (Intercostals)
- Tighten and relax to control air flow in and out of the lungs (Bronchial tubes)
-

INTERIOR AND EXTERIOR OF THE HEART

You can see:

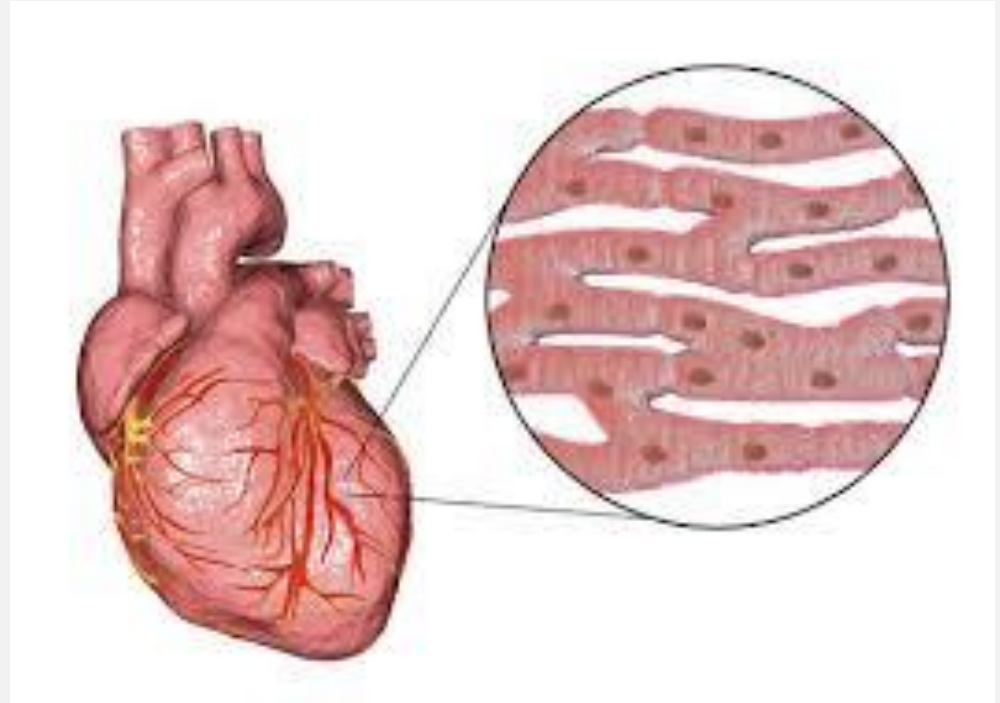
- **Muscle** of the heart and blood vessels (inside and outside)
- **Arteries:** Blood vessels that take blood away from the heart (inside and outside)
- **Veins:** Blood vessels that take blood into the heart (inside and outside)
- **Four Chambers (inside)** - to hold blood; to contract and relax to move blood
- **Valves (inside)** - to keep blood moving in one direction



<https://www.bioedonline.org/lessons-and-more/lessons-by-topic/nature-of-science/scientific-decision-making/heart-and-circulation-outside-and-inside/>

MUSCULAR AND CIRCULATORY CONNECTIONS

- The heart and blood vessels are made of muscle
- Heart muscle is a thick, strong muscle allowing the heart to beat non-stop without getting tired
- Muscle of the blood vessels and heart help our bodies maintain proper blood flow and blood pressure (**circulatory and all other organ systems**)
- Muscles help us to open and close our airways (**respiratory and circulatory**)



<https://stock.adobe.com/ca/search/images?k=cardiac+muscle>

THE CARDIAC CYCLE CONNECTIONS

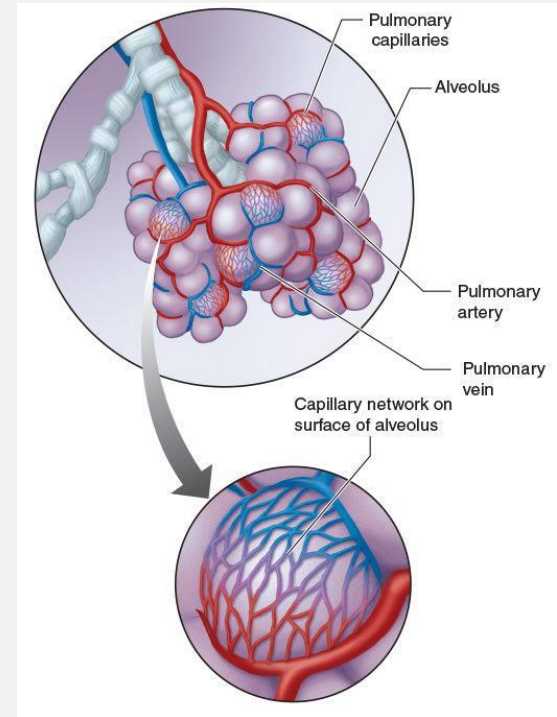
- The heart and blood vessels are made of muscle (**Muscular, Circulatory**)
- The **cardiac cycle** refers to one full **contraction** and **relaxation** phase of the heart (one full heartbeat) - this would not be possible if the heart was not made of **muscle** (**Muscular, Circulatory**)
- During the **cardiac cycle** the blood is moved around the body
- Some blood moves to the lungs to get oxygen, while some blood moves to the organs to give away oxygen so the organs can keep functioning (**Respiratory, Circulatory and all other organ systems**)
- The blood vessels of the heart (veins and arteries) are connected by **capillaries** – where veins and arteries meet (**Muscular, Circulatory**)

BLOOD FLOW THROUGH THE HEART: SYSTEM CONNECTIONS

- Inside the lungs the deoxygenated blood gets oxygen by **diffusion** from the **alveoli (respiratory and circulatory systems)**
- This blood is now oxygenated blood that travels to body organs to give them oxygen (that allows them to stay alive and perform functions) by **diffusion (circulatory and all other organ systems)**
- This blood is now deoxygenated blood that re-enters the heart
- The cardiac cycle starts again

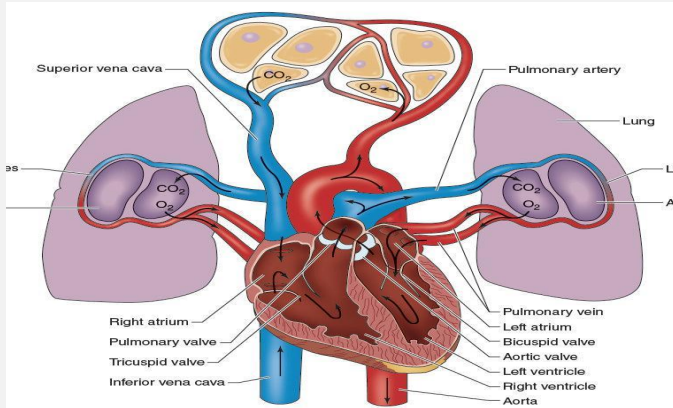
RESPIRATORY CONNECTIONS: CELLULAR RESPIRATION

- This picture shows an up-close view of the **alveoli (moist air sacs that hang at the end of the bronchial tubes)** and their surrounding blood vessels
- Oxygen **diffuses** into the bloodstream from the alveoli (**respiratory, circulatory**)
- Oxygen travels around the body on red blood cells diffusing into organs (via the capillaries) that need oxygen to stay alive (**circulatory, respiratory, muscular**)
- Nutrients, containing glucose (a sugar), **diffuse** into the bloodstream from the food we eat (**respiratory, circulatory and digestive**)
- The glucose and oxygen meet in the bloodstream and undergo a **chemical change** called **cellular respiration** that produces carbon dioxide, water, and energy (**respiratory, circulatory and digestive**)

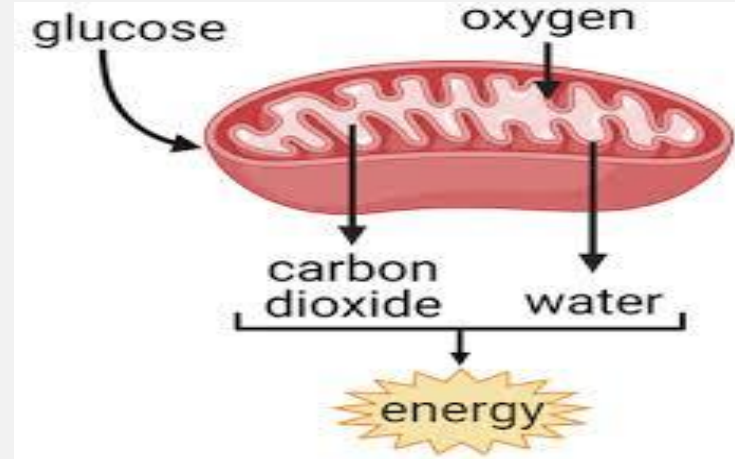


DIAGRAMS

This diagram shows connections between the heart and the respiratory system



This diagram shows the starting materials and new end products of cellular respiration



<https://canada.humankinetics.com/blogs/excerpt/nerves-arteries-and-veins-of-the-heart-and-lungs>

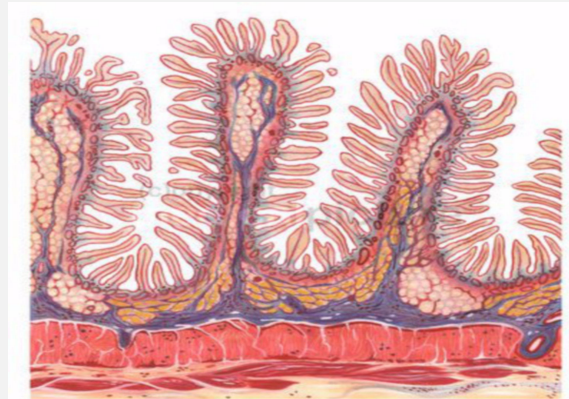
CONNECTIONS: SURFACE AREA

- Surface area is important in our bodies
- **Surface area** is the area or space on the outside of an object
- The shape of the alveoli increases surface area for gas exchange with the bloodstream (**circulatory and respiratory**)
- The small intestine (in humans) and the intestine in worms, along with the inner lining of the stomach are folded (**different digestive systems**)
- Folds give the organ more surface area for nutrient absorption

This picture shows the villi (or folds) in the small intestine of humans

Surface area is a connection between the following systems:

- Digestive and respiratory
- Different digestive systems: worms and humans



CONNECTIONS: DIFFUSION

- **Diffusion is** the passive movement of molecules from an area of high concentration to an area of low concentration
- **Diffusion** allows oxygen to pass from the alveoli into the bloodstream (**circulatory and respiratory**)
- **Diffusion** allows carbon dioxide to pass from the bloodstream into the alveoli so it can exit via our respiratory tract (**circulatory and respiratory**)
- **Diffusion** allows nutrients to pass from the small intestine and stomach into the bloodstream (**digestive and circulatory**)